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Photobiomodulation preconditioning for oral mucositis prevention and quality of life improvement in chemotherapy patients: a randomized clinical trial

Marwa Khalil[1*], Omar Hamadah[1] and Maher Saifo[2,3]

Abstract

Background Given the suffering experienced by cancer patients, effective solutions must be found to prevent painful and debilitating side effects of anti-cancer treatment. This trial aims to study the effect of preconditioning with photobiomodulation in preventing oral mucositis and xerostomia in cancer patients undergoing chemotherapy alone for the first time, and to examine its role in improving patients' quality of life.

Materials and methods This is a prospective, randomized, double-blind clinical trial including 45 patients divided into three age- and sex-matched groups. Group 1 received basic oral care instructions before undergoing chemotherapy. Group 2 received basic oral care instructions plus photobiomodulation using an intraoral 650 nm diode laser. Group 3 received basic oral care instructions plus photobiomodulation using a 650 nm diode laser intraorally and a 980 nm diode laser extraorally.

Results After one week of follow-up, 73.3% of patients in Group 2 and 80% in Group 3 did not develop oral mucositis. The remaining patients in both groups experienced only mild erythema. In contrast, all patients in Group 1 developed oral mucositis, ranging from mild erythema to ulceration greater than 3 cm^2. After two weeks of follow-up, 80% of patients in Group 2 and 86.7% in Group 3 remained free of oral mucositis. The remaining patients in both groups experienced only mild erythema. In contrast, 93.3% of patients in Group 1 developed oral mucositis, with cases ranging from mild erythema to ulcers larger than 3 cm^2. There were statistically significant differences between the three groups in oral mucositis assessment scale after one week and after two weeks ($p < 0.001$). Specifically, ulcerative mucositis was not observed in both laser groups, while it was observed in 13.3% of patients in Group 1. However, these differences were not statistically significant ($p = 0.129$). Regarding Late Effects in Normal Tissue (LENT-SOMA) scale, there was a statistically significant difference between the three groups studied ($p = 0.037$). There was also a statistically significant difference in the Oral Health Impact Profile (OHIP-14) between the three groups studied ($p = 0.003$ after one week, $p = 0.023$ after two weeks).

*Correspondence:
Marwa Khalil
marwa.khalil@damascusuniversity.edu.sy

Full list of author information is available at the end of the article

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Conclusion Preconditioning before starting chemotherapy, whether with the intraoral red laser alone or in combination with the extraoral infrared laser, has shown significant results in preventing oral mucositis and dry mouth, and it has also played a major role in improving patients' quality of life.

Trial registration This trial was registered in ISRCTN registry under no ISRCTN70634383 (https://doi.org/10.1186/ISRCTN70634383) on 24/07/2023.

Keywords Photobiomodulation, Preconditioning, Oral mucositis, Xerostomia, Quality of life

Background

The incidence of stomatitis increased dramatically with the introduction of chemotherapy in the 1940s. In 2007, the term "mucositis" was introduced to describe lesions resulting from the cytotoxic effects of radiation and/or chemotherapy, following the discovery of the complex mechanisms underlying stomatitis [1].

The occurrence and severity of oral mucositis (OM) are primarily influenced by the chemotherapy regimen and dosages. For instance, the incidence of OM can reach 90% in patients treated with 5-fluorouracil [2], 75% with docetaxel and capecitabine, 70% with pralatrexate, 98% (grade 3 and 4) with cyclophosphamide and etoposide [3], 86% with melphalan [4], and 50% with high doses of methotrexate [5].

Chemotherapy is also associated with salivary gland dysfunction. This condition is typically temporary and reversible; however, it can cause pain, interfere with speech, and make chewing difficult. The quantity and quality of the patient's saliva are often altered as a result [6]. Chemotherapy drugs such as fluorouracil, cyclophosphamide, and adriamycin can impair salivary flow, leading to taste changes and a burning sensation. Furthermore, the reduction in lysozyme, lactoferrin, immunoglobulins, and other antibacterial substances further compromises the body's defense mechanisms [7].

Photobiomodulation therapy (PBM) involves applying light in the 600–1000 nm wavelength range to damaged or potentially damaged tissue. The goals of PBM include pain relief, improved healing, and reduced inflammation [8, 9].

The use of intraoral PBM treatment with laser for the prevention of OM and related discomfort was recommended by the Multinational Society for Supportive Care in Cancer and the International Society of Oral Oncology (MASCC/ISOO) expert panel. These recommendations were specifically limited to patients undergoing hematopoietic stem cell transplantation and those with head and neck cancers receiving radiotherapy. Nevertheless, recommendations for other patient groups or different light sources were not provided [10].

Currently, there are no effective treatments or standardized guidelines for preventing oral side effects caused by anticancer chemotherapy. This lack of effective prevention negatively impacts patients' quality of life, prognosis, and care requirements [11].

The use of PBM for the treatment or prevention of OM is still in its early stages and faces several challenges in clinical practice. There was a significant variance in the technical parameters [12].

Therefore, we conducted this study to compare the effects of two clinical protocols: intraoral red laser versus extraoral infrared laser combined with intraoral red laser. Our aim was to evaluate their effectiveness in preventing oral mucositis and dry mouth in cancer patients undergoing chemotherapy for the first time, and to assess their impact on the quality of patients' lives.

Material methods

Trial registration

This trial was registered in ISRCTN registry under no ISRCTN70634383 (https://doi.org/10.1186/ISRCTN70634383) on 24/07/2023.

Ethics approval(s)

Approved 18/01/2023, Scientific Research Council at Damascus University (Damascus University, Damascus, 00963, Syria; +963 11 33923000; president@damasuniv.edu.sy), ref: 2027.

Study design

Prospective randomized controlled double-blind study.

Participant inclusion criteria

Cancer patients undergoing their first session of chemotherapy at Albairouni Hospital in Dmascus, Syria, who meet the following criteria, were included:

- Digestive tract cancer patients.
- Chemotherapy regimens: FOLFOX (leucovorin calcium, fluorouracil, and oxaliplatin) or XELOX (oxaliplatin and capecitabine).
- Neutrophil count ≥ 1500 cells/μL.
- Platelet count ≥ 100,000/μL.
- A healthy oral mucosa.
- Karnofsky performance status index: > 60 [13].

Exclusion criteria

- Radiotherapy in the head and neck area.
- Malignant or potentially malignant lesions of the oral cavity.
- Oral infections.
- Oral bleeding.
- Diabetes.
- Undergoing any other procedures to prevent oral mucositis.
- Patients unable to commit to the study.

Sample size calculation

The minimum sample size was calculated based on a previous study (Arbabi-Kalati, 2013) [14] using the website OpenEpi (www.openepi.com). Taking into account a significance level (1 - α) of 95%, a study power (1 - β) of 80%, and a ratio of 1:1 (unexposed/exposed), the percentage of patients exposed to photobiomodulation therapy in that study who developed grade II or more severe mucositis was 8.33%, while the percentage in the unexposed (control) group was 91.6% (odds ratio: OR = 0.008). Accordingly, the minimum sample size required was 7 patients per group.

Our study included 45 subjects, distributed into three groups.

Randomization

After achieving acceptance criteria, randomization was performed using the online software www.graphpad.com/quickcalcs/randomize1.cfm.

Blinding

Only the researcher responsible for administering the procedures knew which procedure was assigned to each patient. The patients were blinded to the type of procedure they received. The procedures were identical across all three groups, with the laser treatment simulated using the same laser device, in the same manner and for the same duration, but with the power button turned off. The researcher implementing the procedures was also blinded to the outcomes, as the patients were examined by independent evaluators.

Intervention

Group 1: Before the start of the first chemotherapy session, informed consent was obtained from all participants included in the study. Basic oral care instructions were provided both verbally and in writing. These instructions included brushing teeth for 90 s using a soft toothbrush, using dental floss, rinsing the mouth with sterile water, drinking plenty of water, brushing teeth after meals, and avoiding alcohol, cigarette smoking, hot or cold beverages, spicy, sour, or hard foods. If the patient was wearing a full removable prosthesis, they were instructed to brush and rinse their mouths twice a day and to remove the prosthesis at bedtime [15].

Group 2: Patients received the same basic oral care instructions as those in Group 1, in addition to intraoral photobiomodulation with a diode laser (TRIPLO MULTI-WAVE DIODE LASER, MEDENCY®): 635 nm; power 100 mW, energy density 4 J/cm² (Table 1). The number of irradiation points was as follows: 8 on the buccal mucosa (4 on each side), 4 on the labial mucosa (2 upper and 2 lower), 8 on the tongue (3 on the right side, 3 on the left side, and 2 on the ventral surface), 2 on the floor of the mouth, and 2 on the soft palate (Fig. 1). This was performed before the start of the first chemotherapy session (on the same day).

Group 3: Patients received basic oral care instructions along with intraoral photobiomodulation using the same parameters and irradiation points as in Group 2. This was followed by extraoral photobiomodulation with a diode laser: wavelength 980 nm; power 100 mW, energy density 4 J/cm² (Table 1), with 6 irradiation points as shown in (Fig. 2). This procedure was performed before the start of the first chemotherapy session (on the same day).

Primary outcome measure

The presence and severity of oral mucositis were measured using the Oral Mucositis Assessment Scale (OMAS) (Table 2) at 7 and 14 days after the start of the first chemotherapy session [17].

The OMAS score is obtained by summing the erythema and ulceration/pseudomembrane subscores at each site (possible score range, 0 to 5), and then averaging these scores across all sites (The upper and lower lips, right and left buccal mucosa, ventral/lateral tongue, floor of the mouth, and soft and hard palates) [17].

Table 1 Laser parameters used

	Wavelength nm	Power Density mW/cm²	Time per Spot, s	Energy Density J/cm²	Spot Size cm²	No. of Sites	Duration
Intraoral PBM	**635**	**200**	**20**	**4**	**0.5**	**24**	Before the start of the first chemotherapy session (on the same day)
Extraoral PBM	**980**	**200**	**20**	**4**	**0.5**	**6**	Before the start of the first chemotherapy session (on the same day)

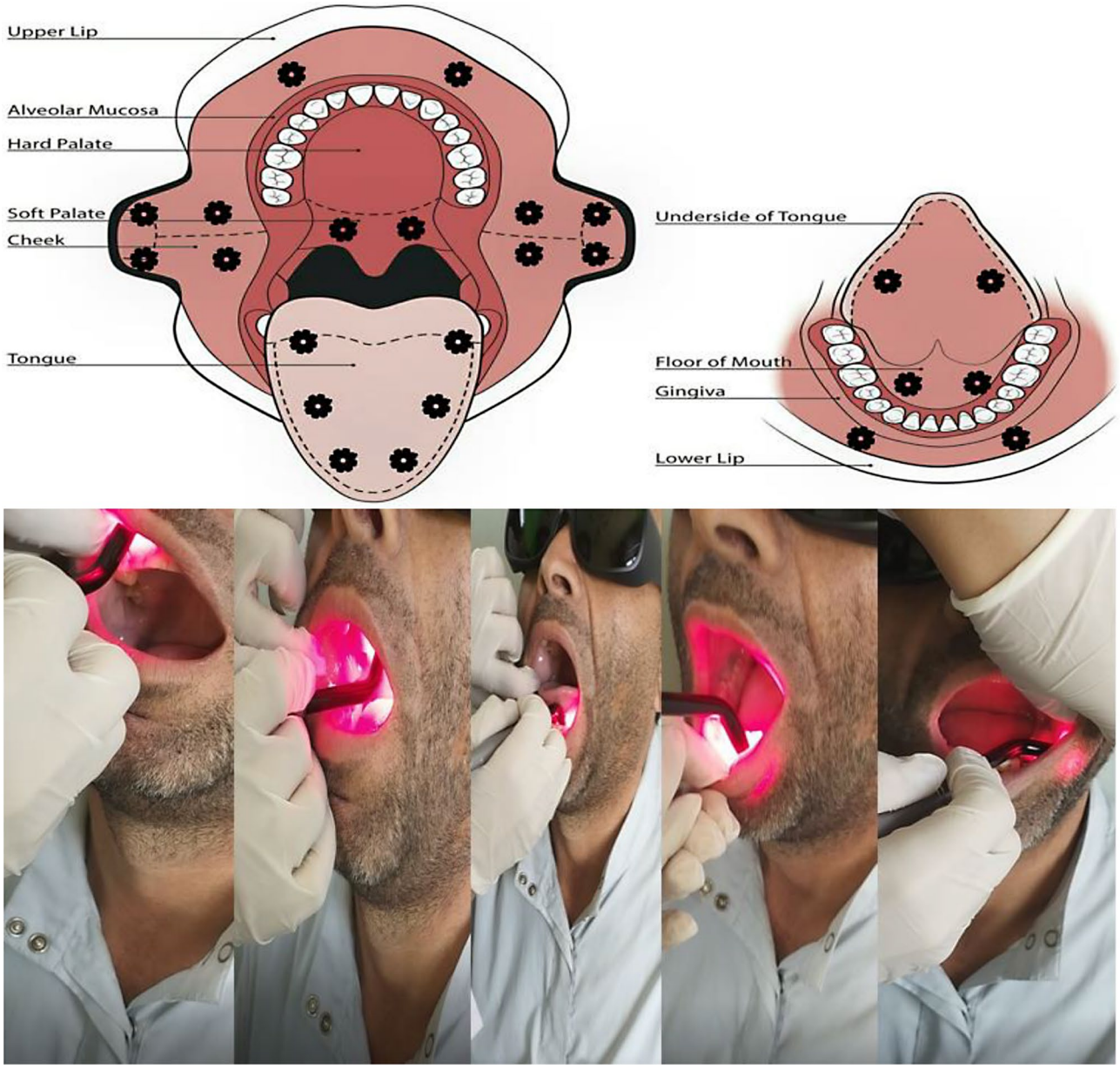


Fig. 1 Intraoral application of PBM

Secondary outcome measures

Secondary outcome measures were assessed at baseline, and at 7 and 14 days:

Quality of life was measured using the Oral Health Impact Profile (OHIP-14) [18]. Patients answered 14 questions within the questionnaire, assigning values to their answers: 0 (never), 1 (almost never), 2 (sometimes), 3 (often), and 4 (always). The points were then summed (0–56), where 56 indicates the greatest impact on quality of life, and the results were divided into seven domains: functional limitation, physical pain, psychological discomfort, physical disability, psychological disability, social disability, and handicaps.

- Xerostomia assessment was based on visual inspection of the oral cavity using the objective grades of Late Effects in Normal Tissue (LENT-SOMA) scale: (Grade 1: Natural moisture, Grade 2: Scant saliva, Grade 3: Absence of moisture; sticky, viscous saliva, Grade 4: Absence of moisture; coated mucosa) [14].

Statistical study

A chi-square test was performed to compare the clinical and demographic characteristics of participants across the three groups.



Fig. 2 Extraoral irradiation points [16]

Table 2 Oral Mucositis Assessment Scale

Erythema	
0 =	no sites with erythema
1 =	mild erythema
2 =	severe erythema
Ulceration/Pseudomembrane	
0 =	no sites with ulceration/pseudomembrane
1 =	ulceration < 1 cm^2
2 =	ulceration 1 cm^2– 3cm^2
3 =	ulceration > 3 cm^2

The Kolmogorov-Smirnov test was used to assess the normal distribution of the values for each of the studied parametric variables, followed by a one-way analysis of variance (ANOVA) to compare the means between the three groups at each time point separately. For pairwise comparisons, Student's t-test was used.

For ordinal variables, the Kruskal-Wallis test was employed to compare the frequency of categories across the three studied groups.

The analyses were performed using SPSS software (version 24). A p-value of less than 0.05 was considered statistically significant.

Results

Our study included 45 patients distributed into three groups (Fig. 3) matched in terms of age, sex, and type of chemotherapy. (Table 3)

Regarding OMAS, there were statistically significant differences between the three groups after one week and after two weeks ($p < 0.001$). (Figs. 4 and 5)

When analyzing each two groups separately, we observed statistically significant differences when comparing the control group with both the intraoral laser group and the intraoral & extraoral laser group. However, no statistically significant differences were found when comparing the intraoral laser group with the intraoral & extraoral laser group.

After one week of follow-up, 73.3% of patients in the intraoral laser group showed no signs of oral mucositis, while 26.7% exhibited only mild erythema. In the intraoral and extraoral laser group, 80% of patients remained free of mucositis, with the remaining 20% experiencing mild erythema. In contrast, all patients in the control group developed oral mucositis, ranging from mild erythema to ulceration exceeding 3 cm^2 (Fig. 6). After two

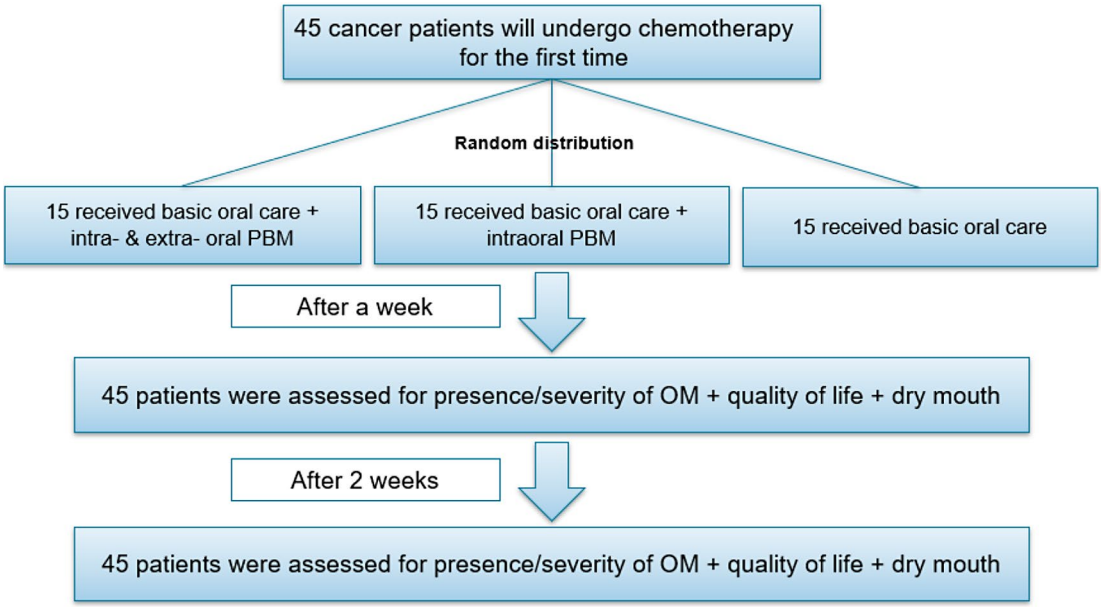


Fig. 3 Flowchart of study participants

Table 3 Distribution of clinical and demographic features of participants among the three groups

	Categories	GROUP 1	GROUP 2	GROUP 3	*P*-value
Age	< 45	2	2	2	1
	> 45	13	13	13	
Gender	Male	9	9	9	1
	Female	6	6	6	
Tumor	Colon Cancer	10	11	11	0.997
	Stomach Cancer	2	2	2	
	Pancreatic Cancer	1	1	1	
	Esophageal cancer	2	1	1	
Drug	FOLFOX	5	5	5	1
	XELOX	10	10	10	
OHIP-14 at baseline	Domain 1	13	13	13	1
	Domain 2	2	2	2	
LENT SOMA at baseline	Grade 1	15	15	15	1

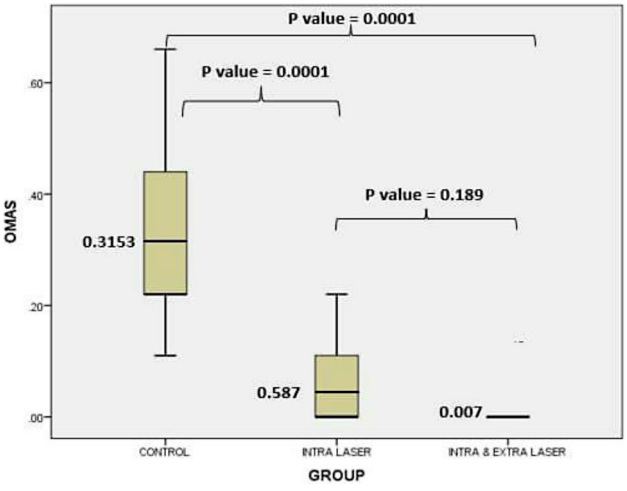


Fig. 4 Means of OMAS values between groups after a week of follow-up

weeks of follow-up, 80% of patients in the intraoral laser group remained unaffected by oral mucositis, while 20% experienced only mild erythema. In the intraoral and extraoral laser group, 86.7% of patients did not develop mucositis, with 13.3% exhibiting mild erythema. Conversely, 93.3% of patients in the control group developed oral mucositis, ranging from mild erythema to ulcers larger than 3 cm^2 (Fig. 7). Ulcerative mucositis was not observed in either laser group at both follow-up periods, whereas it was observed in 13.3% of patients in the control group. However, these differences were not statistically significant ($p = 0.129$).

There was a statistically significant difference in the Oral Health Impact Profile (OHIP-14) between the three groups studied [$p = 0.003$ after one week (Fig. 8), $p = 0.023$ after two weeks (Fig. 9)].

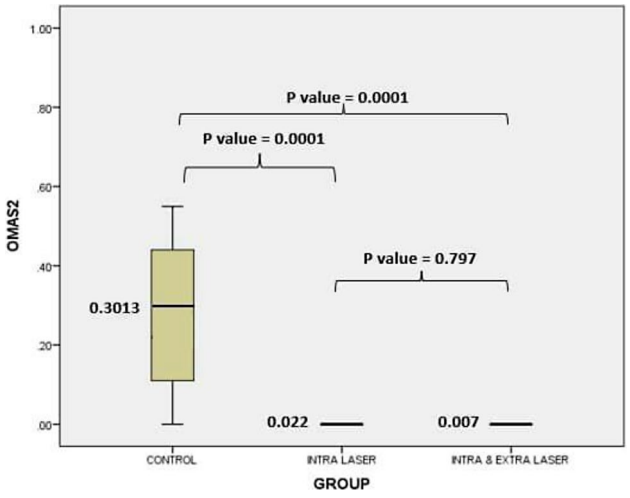


Fig. 5 Means of OMAS values between groups after 2 weeks of follow-up

There was a statistically significant difference in the LENT SOMA scale between the three groups studied ($p = 0.037$). (Figs 10 and 11)

None of the participants developed any side effects resulting from the photobiomodulation treatment.

Discussion

Oral mucositis (OM) is a common and severely debilitating side effect of non-surgical cancer treatments. Painful mouth sores, redness, and damage to the mucosal lining are the hallmarks of OM. OM is often underreported, particularly in patients undergoing chemotherapy [19]. Effective supportive care is crucial for managing oral mucositis, as it typically develops early after chemotherapy, often within one to two days. OM is associated with severe pain, taste alterations, dysphagia, nutritional deficiencies, secondary fungal infections, bacterial infections, and changes in speech. These side effects significantly impact patients' quality of life and may necessitate dose adjustments or early discontinuation of anticancer treatment [20–22].

The most effective strategy for managing mucositis is prevention rather than mitigation or repair. Once lesions develop, management becomes increasingly complex due to the clinical complications they cause and the impaired healing process, which is often exacerbated by systemic debilitation from chemotherapy and the risk of secondary infections. Additionally, even after healing occurs, the affected tissues do not fully return to their pre-mucositis state. This is attributed to residual angiogenesis, which results in weakened tissue that is more susceptible to recurrence [23].

Since ancient times, preconditioning tissues and organs before medical intervention has been a well-established practice. This approach aims to enhance the stress tolerance of cells, including their ability to withstand oxidative

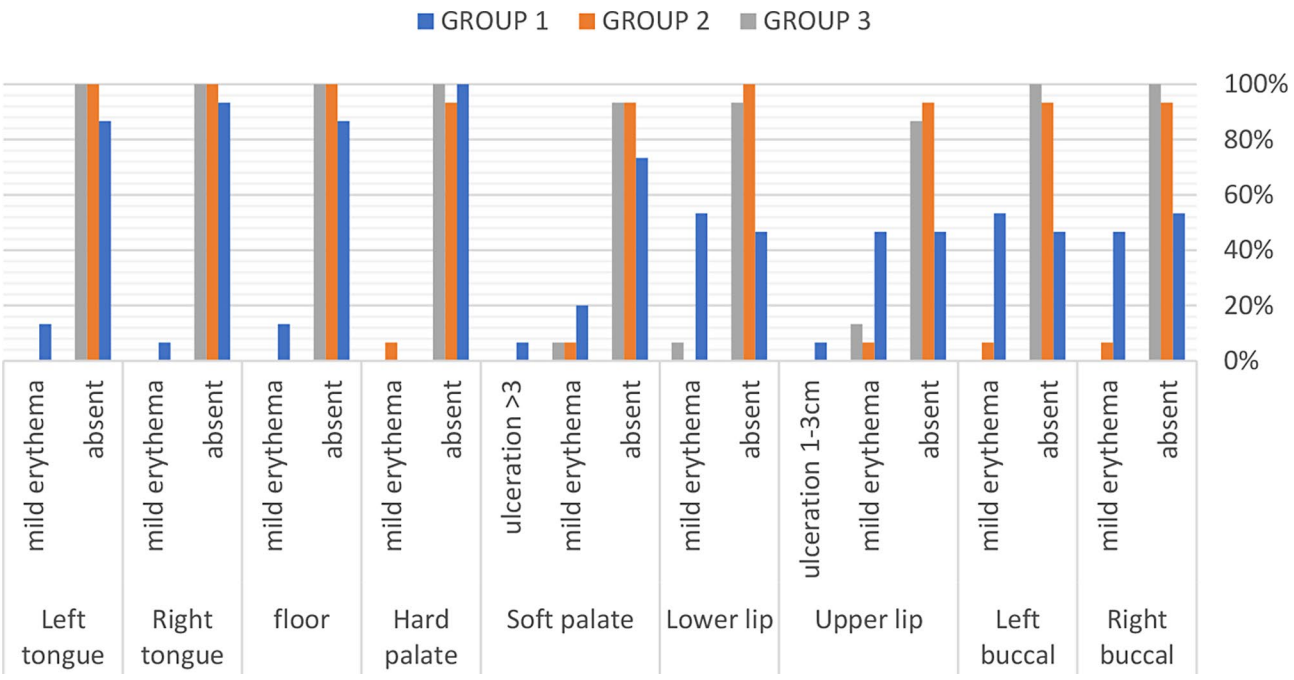


Fig. 6 Percentage of patients experiencing oral mucositis according to the oral site after a week

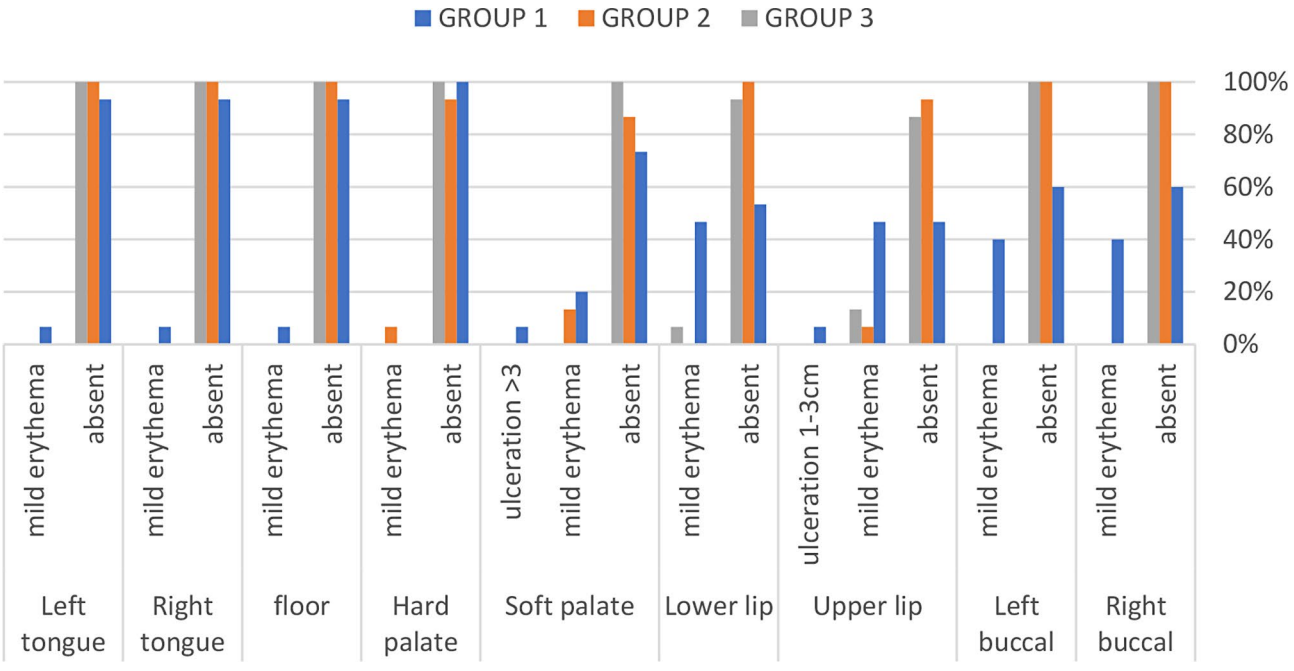


Fig. 7 Percentage of patients experiencing oral mucositis according to the oral site after 2 weeks

stress and treatment-related damage. Research has demonstrated that prior exposure to light can increase cellular flexibility and resistance to chemical and radioactive damage [24–26].

The data supporting the potential therapeutic and preventive effects of photobiomodulation (PBM) on the side effects of cancer therapy was detailed by the World Association of Photobiomodulation Therapy (WALT) [27].

A literature review revealed several trials, the vast majority of which assessed the efficacy of PBM in patients with head and neck cancer receiving radiotherapy or in those undergoing hematopoietic stem cell transplantation (HSCT). This focus persists despite the fact that certain chemotherapy regimens—such as 5-fluorouracil, docetaxel in combination with capecitabine, pralatrexate, and cyclophosphamide with etoposide—can induce

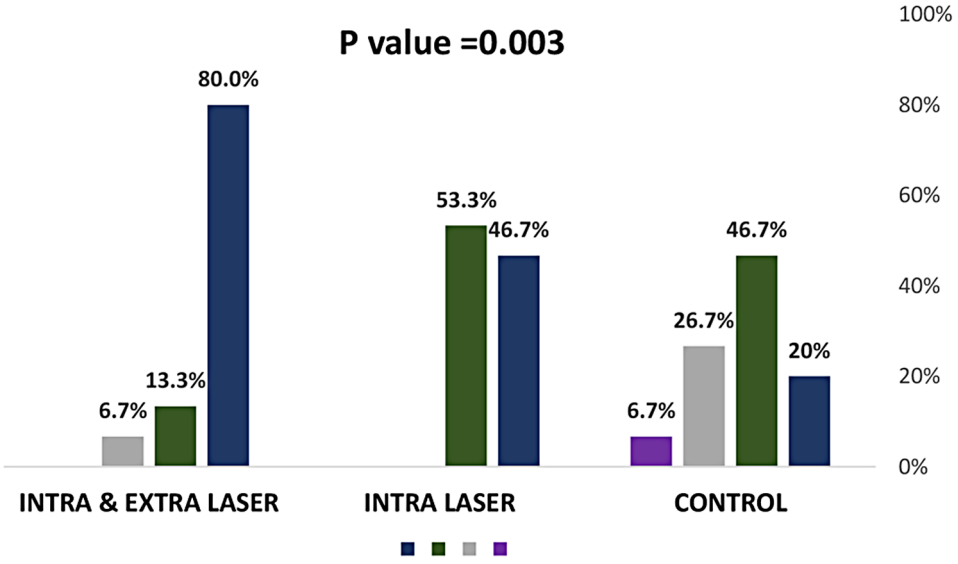


Fig. 8 Distribution of OHIP-14 domain percentages among the three groups after one week of follow-up

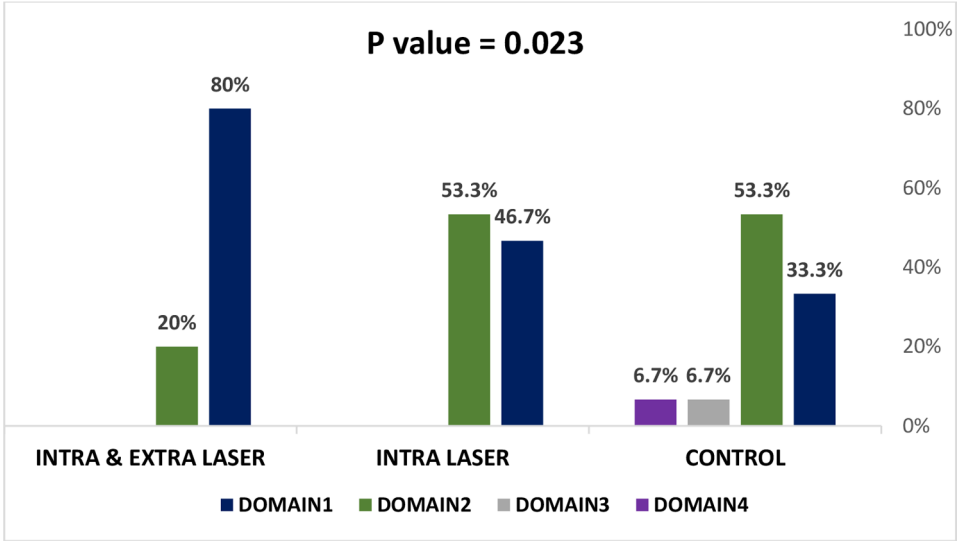


Fig. 9 Distribution of OHIP-14 domain percentages among the three groups after two weeks of follow-up

mucositis in a significant number of patients. Moreover, the incidence rates of mucositis in chemotherapy patients are comparable and very close to those observed in patients with head and neck cancer and HSCT recipients [28–33].

In the current study, we applied PBM before the start of the chemotherapy session, which aligns with WALT recommendations to administer PBM within 30–120 min prior to tumor treatment. We used a 635 nm wavelength for intraoral irradiation, consistent with WALT recommendations to use an LED/laser device with a visible wavelength (630–680 nm) for the prevention of mucositis [27]. The power settings were within the range suggested by Bensadoun et al. [34] The energy density employed was within the range recommended by Cronshaw et al. in their systematic review, which advocates using a lower energy density (2–5 J/cm^2) for the prevention and healing of lesions, and a higher energy density (10–15 J/cm^2) for OM-associated analgesia rather than biostimulation [26].

We used the Oral Mucositis Assessment Scale (OMAS) to evaluate the presence and severity of oral mucositis because it is an objective, quantitative measure suitable for research purposes. The OMAS primarily relies on measuring the dimensions of ulceration and erythema in nine different locations within the oral cavity [17].

According to our study, the incidence of mucositis was 100% in the first group, which appears to be higher than the rates reported in the literature [2]. This discrepancy may be attributed to the fact that 86.7% of our patients presented with mild erythema, a degree of oral mucositis

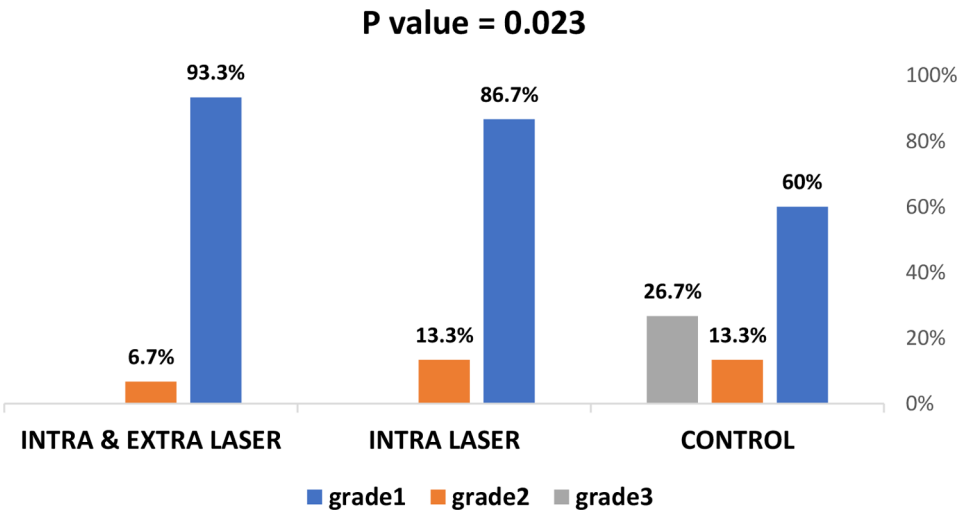


Fig. 10 The percentage distribution of LENT SOMA grades among the three groups after a week of follow-up

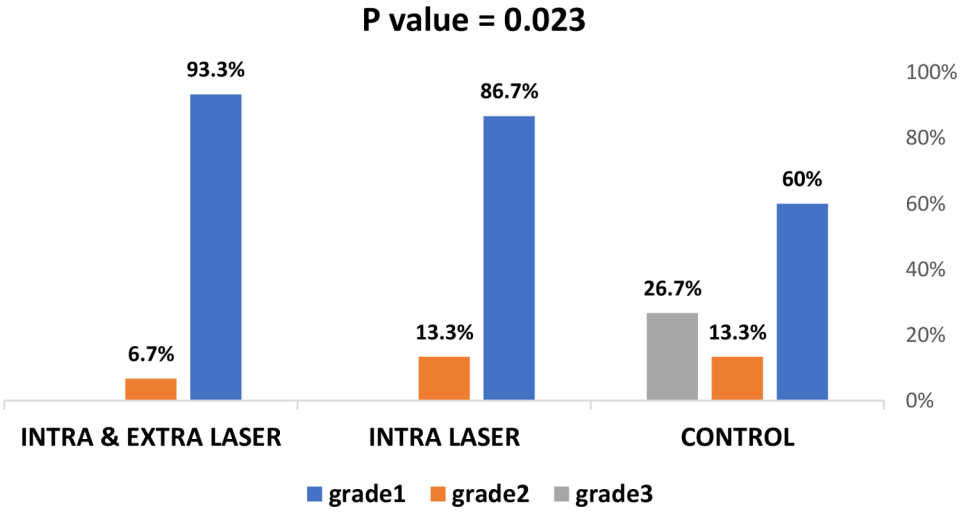


Fig. 11 The percentage distribution of LENT SOMA grades among the three groups after two weeks of follow-up

that is typically not reported by patients but is detected during clinical examination by the clinician. It is important to consider that most incidence rates of oral mucositis are underestimated because patients tend to report it only when it reaches more severe degrees.

Through our review of the literature, we identified six studies evaluating the role of Photobiomodulation (PBM) in preventing oral mucositis. All these studies confirmed the positive impact of PBM, except for one study by Cruz et al., which found no benefit from laser application. Our study supports the significant role of photobiomodulation in preventing oral mucositis. We concur with most previous studies, including the research by Arbabi-Kalati et al. [14], who used a laser with the following parameters: 630 nm, 30 mW, and 5 J/cm^2, applied to ten sites within the oral cavity of patients with hematological and solid tumors. Likewise, Rozza-de-Menezes et al., who applied the intraoral laser to patients with solid tumors undergoing chemotherapy with fluorouracil and/or doxorubicin according to the following parameters: 660 nm, 4 J/cm^2 and 50 mW, found that PBM has a noticeable role in preventing OM, with patients showing only 75% erythema [35]. There was also alignment with the results of de Castro et al., who compared the efficacy of a 660-nm red laser and an 830-nm infrared laser for the prevention and treatment of oral mucositis in children receiving high-dose methotrexate for acute lymphoblastic leukemia and found that prophylactic laser therapy resulted in better outcomes compared to patients who did not receive any preventive intervention [36]. Similarly, the Malta study observed a lower incidence of oral mucositis in the PBM group, which used 660-nm and 808-nm lasers, among stage II-IV breast cancer patients undergoing treatment with doxorubicin and cyclophosphamide [37] The positive role of PBM in the prevention of oral mucositis, as observed in our study and most other

studies, is due to its ability to favorably influence cellular metabolism. This occurs when light energy is transferred to the mitochondria, leading to an increased production of adenosine triphosphate (ATP) and nitric oxide, which subsequently triggers complex effects on gene expression. These effects result in many beneficial changes, such as enhanced production of collagen, growth factors, and an increased rate of cell division, all of which contribute to the healing process. Additionally, PBM plays a role in eliminating mediators of acute inflammation and modulates cytoplasmic reactive oxygen species, thus enhancing the immune response in targeted areas [38, 39].

Our only disagreement concerns the study by Cruz et al., which attributed the ineffectiveness of PBM to the increased susceptibility of children to mucositis. This contrasts with the numerous studies conducted in pediatric populations that have demonstrated positive outcomes of PBM in preventing mucositis [40–42]. A more plausible explanation for the findings of Cruz et al. is that some participants in their study had previously developed mucositis due to prior chemotherapy regimens, which may have rendered their mucosa more susceptible to further damage because of residual blood vessel formation. This highlights the importance of administering PBM to healthy tissue before the initiation of any anticancer therapy [43].

Several studies have demonstrated the superiority of red light in preventing mucositis, including the study by de Castro et al., which compared red light with infrared light in children with leukemia receiving high doses of methotrexate [36], as well as the study by Schubert et al. on hematopoietic stem cell transplantation patients [44]. However, this does not preclude the use of infrared light, which can be effectively combined with red light due to its deeper penetration. This allows infrared light to reach difficult-to-access areas, such as the oropharyngeal region, helping to prevent lesion development and thereby improving swallowing and quality of life [16]. For the above-mentioned reasons, we combined infrared with red light. In fact, the proportion of patients without OM in the infrared laser plus red laser group was higher compared to the intraoral red laser alone group, but the differences were not statistically significant. This may be attributed to the fact that extraoral infrared laser has a positive effect in preventing OM but the greater effect is due to intraoral red laser. We agree with the study of Malta et al., which shared red light with infrared light in breast cancer patients undergoing chemotherapy with doxorubicin and cyclophosphamide and found that the incidence of OM was significantly reduced in these patients [37].

Chemotherapy also causes dysfunction of the salivary glands by inhibiting the lubricating, moisturizing, and antibacterial activities of saliva. Chemotherapy appears to have different effects on basal and stimulated saliva. While the basal pH of saliva remains unchanged, the pH of stimulated saliva becomes acidic. This acidity, combined with decreased saliva flow, leads to changes in beneficial flora. In addition, electrolyte modifications have been observed. During ion rearrangement, increases in Na^+ and K^+ disrupt salivary duct transport systems, causing damage [6, 7]. So far, there is no conclusive evidence on the effect of PBM on the salivary glands, but many studies and reviews have suggested the benefit of applying PBM to enhance the function of the salivary glands. For example, Golez et al. highlighted the role of the laser in alleviating dry mouth and hyposalivation [45]. Similarly, Sousa et al. observed an increase in the rate of saliva production in patients undergoing PBM [46]. In our study, we observed that PBM significantly reduced the incidence of xerostomia in patients undergoing chemotherapy, which is consistent with the findings of Arbabi Kalati et al. [14] The most likely explanation for this effect may be the role of PBM in stimulating the submandibular gland, which is responsible for producing the majority of saliva (about 70%) in the unstimulated state [47]. However, there was no significant difference between the two PBM protocols applied.

Several studies have evaluated the impact of PBM on patients' quality of life. For instance, Gautam et al. studied patients with head and neck cancer undergoing combined chemotherapy and radiation, and found that PBM was effective in improving patients' subjective experiences of OM and overall quality of life [48]. Likewise, Silva et al. found that PBM improved quality of life in patients with head and neck cancer treated with radiotherapy [49]. In our study, we observed that the Oral Health Impact Profile (OHIP-14) scores were better in the PBM group, whether intraoral alone or combined with extraoral application. This improvement is attributed to PBM's role in reducing the incidence of OM and dry mouth, which subsequently alleviates symptoms such as pain, difficulty eating, dysphagia, taste changes, and other physical and psychological consequences that affect patients' quality of life. These findings are consistent with the study by Malta et al., which demonstrated that PBM enhanced overall health status and quality of life in patients with breast cancer undergoing chemotherapy [37].

Conclusion

Preconditioning before starting chemotherapy, whether with the intraoral red laser alone or in combination with the extraoral infrared laser, has shown significant results in preventing oral mucositis and dry mouth, and it has also played a major role in improving the quality of life of patients.

Limitations

A limitation of this study is the short follow-up period, as we were unable to monitor patients for longer durations. Furthermore, extending the follow-up could introduce potential confounding factors, as patients who develop side effects may undergo various treatments, such as pain management, which could impact the accurate assessment of the true effectiveness of photobiomodulation therapy in preventing oral complications.

Supplementary Information

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Supplementary Material 1

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Author contributions
M.K contributed to the study design, literature search, clinical studies, data collection and manuscript writing. O.H and M.S contributed to the study design, analysis, interpretation of results and manuscript review. All authors read and approved the final manuscript.

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Data availability
The dataset used during the study are available from the corresponding author upon request.

Declarations

Ethical approval and informed consent to participate:
Ethical Approval was obtained from Scientific Research Council at Damascus University (Damascus University, Damascus, 00963, Syria; +963 11 33923000; president@damasuniv.edu.sy), ref: 2027 on 18/01/2023. Also, all participants in this study provided a signed, informed consent.

Competing interests
The authors declare no competing interests.

Consent for publication
The patients have given their consent for their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

Competing interests
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Author details
[1]Department of Oral Medicine, Faculty of Dentistry, Damascus University, Damascus, Syrian Arab Republic
[2]Department of Medical Oncology, Faculty of Medicine, Damascus University, Damascus, Syrian Arab Republic
[3]Albairouni University Hospital, Damascus University, Damascus, Syrian Arab Republic

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References

1. Pulito C, et al. Oral mucositis: the hidden side of cancer therapy. J Experimental Clin cancer Res. 2020;39:1–15.
2. Curra M, et al. Chemotherapy protocols and incidence of oral mucositis. An integrative review. Einstein (Sao Paulo). 2018;16:eRW4007.
3. Velcheti V, Punekar SR. Handbook of cancer treatment-related symptoms and toxicities E-book. Elsevier Health Sciences; 2021.
4. Blijlevens N, et al. Prospective oral mucositis audit: oral mucositis in patients receiving high-dose melphalan or BEAM conditioning chemotherapy—European blood and Marrow Transplantation Mucositis Advisory Group. J Clin Oncol. 2008;26(9):1519–25.
5. Guimaraes DM, et al. Low-level laser or LED photobiomodulation on oral mucositis in pediatric patients under high doses of methotrexate: prospective, randomized, controlled trial. Support Care Cancer. 2021;29(11):6441–7.
6. Chaveli López B, Gavaldá C, Esteve. and M.G. Sarrión Pérez, *Dental treatment considerations in the chemotherapy patient*. 2011.
7. Mazzeo M-A et al. *Oral signs of intravenous chemotherapy with 5-Fluorouracil and Leucovorin calcium in colon cancer treatment*. 2009.
8. Chung H, et al. The nuts and bolts of low-level laser (light) therapy. Ann Biomed Eng. 2012;40:516–33.
9. de Freitas LF, Hamblin MR. Proposed mechanisms of photobiomodulation or low-level light therapy. IEEE J Sel Top Quantum Electron. 2016;22(3):348–64.
10. Elad S, et al. MASCC/ISOO clinical practice guidelines for the management of mucositis secondary to cancer therapy. Cancer. 2020;126(19):4423–31.
11. Khalil M, Hamadah O, Saifo M. Preconditioning with Photobiomodulation as an effective method in preventing Chemotherapy-Induced oral mucositis: a systematic review. Photobiomodulation Photomed Laser Surg. 2023;41(11):597–607.
12. Abdalla-Aslan R, et al. Clinical use of photobiomodulation for the prevention and treatment of oral mucositis: the real-life experience of MASCC/ISOO members. Support Care Cancer. 2023;31(8):481.
13. Ferrucci L, et al. *Disability, functional status, and activities of daily living*, in *Encyclopedia of gerontology*. Elsevier Inc; 2010. pp. 427–36.
14. Arbabi-Kalati F, Arbabi-Kalati F, Moridi T. Evaluation of the effect of low level laser on prevention of chemotherapy-induced mucositis. Acta Medica Iranica, 2013: pp. 157–62.
15. Peterson D, et al. Management of oral and gastrointestinal mucosal injury: ESMO Clinical Practice guidelines for diagnosis, treatment, and follow-up. Ann Oncol. 2015;26:v139–51.
16. Soto M, et al. Pilot study on the efficacy of combined intraoral and extraoral low-level laser therapy for prevention of oral mucositis in pediatric patients undergoing hematopoietic stem cell transplantation. Photomed Laser Surg. 2015;33(11):540–6.
17. Niscola P, et al. Mucositis in patients with hematologic malignancies: an overview. Haematologica. 2007;92(2):222–31.
18. de Oliveira BH, Nadanovsky P. Psychometric properties of the Brazilian version of the oral Health Impact Profile–short form. Commun Dent Oral Epidemiol. 2005;33(4):307–14.
19. Freytes S, Villa A. *Oral Mucositis*. Care of Head and Neck Cancer Patients for Dental Hygienists and Dental Therapists, 2023: pp. 205–210.
20. Chan CW, et al. Oral mucositis among Chinese cancer patients receiving chemotherapy: effects and management strategies. Asia-Pac J Clin Oncol. 2021;17(2):e10–7.
21. El Mobadder M, et al. Photobiomodulation therapy in the treatment of oral mucositis, dysphagia, oral dryness, taste alteration, and burning mouth sensation due to cancer therapy: a case series. Int J Environ Res Public Health. 2019;16(22):4505.
22. Ueno T, et al. Effects of an oral mucosa protective formulation on chemotherapy-and/or radiotherapy-induced oral mucositis: a prospective study. BMC Cancer. 2022;22(1):1–9.
23. Basile D, et al. Mucosal injury during anti-cancer treatment: from pathobiology to bedside. Cancers. 2019;11(6):857.
24. Agrawal T et al. *Pre-conditioning with low-level laser (light) therapy: light before the storm*. Dose-Response, 2014. 12(4): p. dose-response. 14–032. Agrawal.
25. Darbar A, Darbar R. *Preconditioning and low level laser therapy and post-treatment management in dental practice*. In *mechanisms for low-light therapy VI*. SPIE; 2011.
26. Cronshaw M, et al. Photobiomodulation and oral mucositis: a systematic review. Dentistry J. 2020;8(3):87.
27. Robijns J, et al. Photobiomodulation therapy in management of cancer therapy-induced side effects: WALT position paper 2022. Front Oncol. 2022;12:927685.

28. Antunes HS, et al. Phase III trial of low-level laser therapy to prevent oral mucositis in head and neck cancer patients treated with concurrent chemoradiation. Radiother Oncol. 2013;109(2):297–302.
29. Antunes HS, et al. Long-term survival of a randomized phase III trial of head and neck cancer patients receiving concurrent chemoradiation therapy with or without low-level laser therapy (LLLT) to prevent oral mucositis. Oral Oncol. 2017;71:11–5.
30. Ferreira B, da Silveira FM, de Orange FA. Low-level laser therapy prevents severe oral mucositis in patients submitted to hematopoietic stem cell transplantation: a randomized clinical trial. Support Care Cancer. 2016;24:1035–42.
31. Franco R, et al. Low-level laser therapy for the treatment of oral Mucositis Induced by hematopoietic stem cell transplantation: a systematic review with Meta-analysis. Medicina. 2023;59(8):1413.
32. Nishi H, et al. Efficacy of low-level laser therapy for oral mucositis in hematologic patients undergoing transplantation: a single-arm prospective study. J Personalized Med. 2023;13(11):1603.
33. Kuhn-Dall'Magro A, et al. Low-level laser therapy in the management of oral mucositis induced by radiotherapy: a randomized double-blind clinical trial. J Contemp Dent Pract. 2022;23(1):31–6.
34. Bensadoun R-J, Nair RG. Low-level laser therapy in the prevention and treatment of cancer therapy-induced mucositis: 2012 state of the art based on literature review and meta-analysis. Curr Opin Oncol. 2012;24(4):363–70.
35. Rozza-de-Menezes R, et al. Behaviour and Prevention of 5'Fluorouracil and Doxorubicin-induced Oral Mucositis in Immunocompetent Patients with Solid Tumors: A Randomised Trial. Volume 16. Oral Health & Preventive Dentistry; 2018. pp. 549–55. 6.
36. de Castro JFL, et al. Low-level laser in prevention and treatment of oral mucositis in pediatric patients with acute lymphoblastic leukemia. Photomed Laser Surg. 2013;31(12):613–8.
37. Malta CEN et al. Photobiomodulation therapy prevents dysgeusia chemotherapy induced in breast cancer women treated with doxorubicin plus cyclophosphamide: a triple-blinded, randomized, placebo-controlled clinical trial. Support Care Cancer, 2022: pp. 1–12.
38. Heiskanen V, Hamblin MR. Photobiomodulation: lasers vs. light emitting diodes? Photochem Photobiol Sci. 2018;17(8):1003–17.
39. Prasad B, et al. Efficacy of LASER photobiomodulation in the management of Cancer Treatment Induced oral mucositis: a systematic review. Annals Romanian Soc Cell Biology. 2021;25(6):5259–78.
40. Ávila-Sánchez C, et al. Impact of a protocol for the prevention and care of oral mucositis in pediatric patients diagnosed with cancer. Gaceta Mexicana De Oncologia. 2017;16(2):96–102.
41. He M, et al. A systematic review and meta-analysis of the effect of low-level laser therapy (LLLT) on chemotherapy-induced oral mucositis in pediatric and young patients. Eur J Pediatrics. 2018;177:7–17.
42. Nunes LFM, et al. Prophylactic photobiomodulation therapy using 660 nm diode laser for oral mucositis in paediatric patients under chemotherapy: 5-year experience from a Brazilian referral service. Lasers Med Sci. 2020;35:1857–66.
43. Cruz LB, et al. Influence of low-energy laser in the prevention of oral mucositis in children with cancer receiving chemotherapy. Volume 48. Pediatric blood & cancer; 2007. pp. 435–40. 4.
44. Schubert MM, et al. A phase III randomized double-blind placebo-controlled clinical trial to determine the efficacy of low level laser therapy for the prevention of oral mucositis in patients undergoing hematopoietic cell transplantation. Support Care Cancer. 2007;15:1145–54.
45. Golež A et al. Effects of low-level light therapy on xerostomia related to hyposalivation: a systematic review and meta-analysis of clinical trials. Lasers Med Sci, 2022: pp. 1–14.
46. Sousa A, et al. Photobiomodulation and salivary glands: a systematic review. Lasers Med Sci. 2020;35:777–88.
47. Grewal JS, Jamal Z, Ryan J. *Anatomy, head and neck, submandibular gland.* 2019.
48. Gautam AP, et al. Effect of low-level laser therapy on patient reported measures of oral mucositis and quality of life in head and neck cancer patients receiving chemoradiotherapy—a randomized controlled trial. Support Care Cancer. 2013;21:1421–8.
49. de Silva Ce. Photobiomodulation for the management of xerostomia and oral mucositis in patients with cancer: a randomized clinical trial. Lasers Med Sci. 2023;38(1):101.

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